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Studierichting: Informatica
Oefeningenreeks 5D nr. 4.14

$f(x) = -x^2 + 4$ en $g(x) = 0$, $x = 0$

$v(x) = f(x) - g(x)$

$v(x) = -x^2 + 4$

$x_{1,2} = \pm 2$

x		-2		2	
$v(x)$	$-$	0	$+$	0	$-$

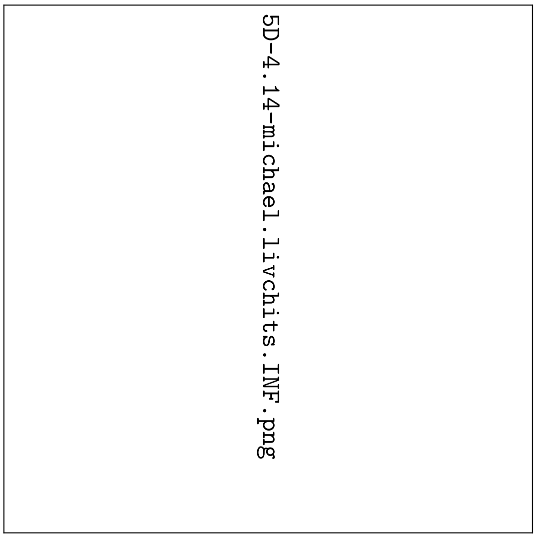
$$I = \int_{-2}^{2} \left(-x^2 + 4\right) dx$$
$$= \int_{-2}^{0} \left(-x^2 + 4\right) dx + \int_{0}^{2} \left(-x^2 + 4\right) dx$$
$$= \left[-\frac{x^3}{3} + 4x\right]_{-2}^{0} + \left[-\frac{x^3}{3} + 4x\right]_{0}^{2}$$
$$= \frac{16}{3} + \frac{16}{3} = \frac{32}{3}$$

Nodig voor de tekening:

$f(x) = -x^2 + 4$ met nulpunten $x = \pm 2$

$f'(x) = -2x$ met nulpunt $x = 0$

x		-2		0		2	
$f(x)$	$-$	0	$+$	$+$	$+$	0	$-$
$f'(x)$	$\nearrow$	$\nearrow$	$\nearrow$	$M(4)$	$\searrow$	$\searrow$	$\searrow$
$f''(x)$	$\frown$	$\frown$	$\frown$	$\smile$	$\frown$	$\frown$	$\frown$



References